CRBCL PLATFORM TECHNICAL ARCHITECTURE & SYSTEM SPECIFICATION

Comprehensive Architectural Inventory, Domain Schema, Integration Models, and Technical Handover Reference

Cowessess First Nation | Chief Red Bear Children's Lodge
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Architecture Baseline: Post-Phase 5 Release Audit

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1. SYSTEM TOPOLOGY & DECOUPLED ARCHITECTURE

The Chief Red Bear Children's Lodge (CRBCL) Family Wellness Case Management Platform is engineered as a modern, decoupled, multi-tier software system. The architecture separates presentation, API orchestration, persistent relational storage, and background processing to guarantee high data availability, sovereign access controls, and horizontal scalability.

• **Web Presentation Tier:** Built with React 18, Vite, and Tailwind CSS. The web client communicates with the backend via a centralized API client implementing automated JWT refresh, unified error envelopes, and client-side route guards.

• **Mobile Presentation Tier:** Cross-platform client built in Flutter / Dart. Provides offline-first capabilities for rural caseworkers using local SQLite caching, local biometric authentication, and an outbox queue engine for bidirectional sync.

• **API & Orchestration Tier:** High-performance asynchronous Python service built on FastAPI and Pydantic v2. Implements dependency-injected database sessions, fine-grained RBAC permission enforcement, and an Authorization-First AI context gateway.

• **Relational Persistence Tier:** PostgreSQL 15+ managed instance utilizing SQLAlchemy 2.0 async engine with connection pooling. Database migrations are rigorously version-controlled through 31 sequential Alembic revision scripts.

2. ARCHITECTURAL EVOLUTION: ORIGINAL SPECIFICATION VS CURRENT IMPLEMENTATION

A rigorous review of the original CRBCL Master Project Document reveals significant positive architectural evolution. Rather than degrading the design, implementation changes were made to enhance First Nations data sovereignty, security, and offline field utility:

System Domain	Original Specification Baseline	Current Production Architecture (0746579)
Database & Schema	Monolithic schema with shared client-case tables	Decoupled domain models across 14 distinct namespaces with versioned audit histories (126 models).

API Architecture	Synchronous Flask/Django endpoints	Asynchronous FastAPI with non-blocking asyncpg connections, Pydantic v2 validation, and OpenAPI 3.1 schemas.
AI / Ask Red Bear	Generic chat interface connected to cloud LLM	Strict Authorization-First AI Gateway, prompt guardrails, field redactions, and simulated stub engine.
Offline Field Sync	Standard web responsive views only	Dedicated native Flutter client with SQLite offline drafting, encrypted storage, and transactional sync engine.
Data Import & Migration	Direct bulk SQL insert from spreadsheets	Isolated quarantine staging wizard, 3-tier duplicate engine, client reconciliation review queue, inactive staff protection.

3. RELATIONAL DATA LAYER & DOMAIN SCHEMA ARCHITECTURE

The relational schema encompasses 126 SQLAlchemy models distributed across 14 business domains. All primary keys utilize RFC 4122 Version 4 UUIDs to prevent ID enumeration attacks and facilitate distributed offline generation. Foreign keys enforce referential integrity with explicit ondelete cascading or restrict rules:

Domain Namespace	Primary Tables & Entities	Relational Integrity & Audit Controls
Identity & Access (RBAC)	users, roles, permissions, role_permissions, user_roles, audit_log	Granular 114 permissions; immutable audit trails on all permission grants.
Person & Client Master	persons, clients, families, person_identifiers, client_approvals	Fuzzy duplicate detection tables; controlled merge history; family linkage graph.
Case Management	cases, case_assignments, case_restrictions, case_links, case_status_history	Mandatory active assignment check; case restriction suppression rules.
Clinical Notes & Evidence	case_notes, case_note_addenda, case_note_attachments, note_persons	Immutable note locking after sign-off; addendum versioning; no hard deletes.
Assessments & Safety	assessments, assessment_sections, questions, answers, safety_plans	Multi-version template engine; supervisor unlock tracking; sequence numbering.
Prevention & Healing	ph_records, ph_intake_links, ph_staff_quarantine, ph_duplicate_queue	3-tier duplicate scoring; provenance lineage; isolated quarantine tables.
Operations, Fleet & Finance	invoices, service_requests, billing_rates, vehicles, fleet_trips, geofences	Multi-fund ledger; strict invoice immutability; vehicle telematics tracking.

4. AUTHENTICATION, RBAC & FIRST NATIONS DATA SOVEREIGNTY

• **Dual Authentication Credentials:** The system supports both standard email/password authentication (Argon2id hashing) and frontline 6-digit secure PIN authentication for rapid, touch-friendly mobile access.

• **Granular 114-Permission RBAC Matrix:** Every endpoint and database action validates discrete permissions (e.g. 'case:create', 'client:medical:read'). Fourteen canonical roles bundle standard permissions.

• **Case-Level Conflict Restrictions:** Caseworkers with personal or familial ties to clients can be restricted from specific cases, automatically blinding them from searches, dashboards, and AI queries.

• **Staff Login Quarantine:** Historical staff records parsed from spreadsheets remain in an INACTIVE quarantine status until formally verified and activated by the System Administrator.

5. ASK RED BEAR AI ARCHITECTURE DEEP-DIVE

Ask Red Bear is engineered around the Gateway and Context Manager patterns to guarantee that AI assistance never bypasses data confidentiality or statutory child welfare rules:

Step 1 — Prompt Guard & Safety Filter: Inbound prompts are scanned for prompt injection attacks or role breakout attempts.

Step 2 — Authorization-First Context Construction: Only records to which the user has explicit, un-restricted access are pulled into the retrieval context.

Step 3 — Allowlisted Tool Execution: Invokes internal deterministic tools (e.g. get_case_summary, get_overdue_goals) rather than allowing hallucinated data.

Step 4 — Field-Level Redaction Engine: Sensitive attributes (medical notes, reporter identities, financial accounts) are stripped before prompt construction.

Step 5 — Speech Transcription as Input Aid: Voice audio is transcribed in memory to the user's text box for explicit human review; audio is never auto-committed as records.

Step 6 — Current LLM Engine State: Currently uses FakeAiProvider in the backend; live production requires provisioning an enterprise API key for a Canadian-hosted sovereign LLM.

6. MOBILE FIELD CLIENT & OFFLINE SYNCHRONIZATION

The Flutter mobile app enables caseworkers to operate in remote reserve areas without cellular connectivity. Offline mutations are written to an encrypted local SQLite outbox table with idempotency keys. Upon reconnecting, the SyncEngine processes the queue sequentially, resolving version conflicts using server-wins timestamps.

7. TECHNICAL DEBT, SYSTEM RISKS & HANDOVER READINESS

• **Staging DNS & CORS Synchronization:** Finalize Railway backend CORS headers to match the Vercel/DigitalOcean frontend domain (https://genserver.online).

• **Third-Party Gateway Setup:** Provide live production credentials for Twilio Canada (SMS alerts), Geotab (telematics), and Anthropic/Azure (AI LLM).

• **End-to-End Browser UAT:** Run automated Playwright/Cypress test runs against live staging to record baseline browser interaction traces.